

Tejas Patil

Davis, CA

tpatil@ucdavis.edu | +1 (530) 979-6396 | [linkedin.com/in/tejas-svpatil](https://www.linkedin.com/in/tejas-svpatil) | github.com/tejassvpatil | [Google Scholar](https://scholar.google.com/citations?user=tejas-svpatil)

Profile

Computer Science PhD student at UC Davis working on multimodal learning, foundation models, in-home sensing, and human-centered software for cognitive health. Production engineering background in Java, Python, TypeScript, APIs, and cloud systems.

Education

University of California, Davis <i>PhD, Computer Science; GPA: 4.0</i> Advisors: Alyssa Weakley and Chen-Nee Chuah	2023–present <i>Davis, CA</i>
University of Pune <i>BE, Computer Engineering; GPA: 3.74</i>	2017–2021 <i>Pune, India</i>

Research and Engineering Experience

UC Davis Health — Graduate Student Researcher <i>Department of Neurology, Everyday Cognition and Resilient Aging Lab</i>	2025–present <i>Davis, CA</i>
- Build remote-care research infrastructure that converts in-home sensor streams into activity summaries with explicit data contracts, access controls, and privacy-aware visualizations. - Developed caregiver–patient messaging, media, calendar, and authorization workflows; refactored Firestore schemas and queries to improve end-to-end responsiveness by approximately 30%. - Co-authored eight peer-reviewed papers on cognitive accessibility, dementia monitoring, LLM delegation, foundation-model activity timelines, and remote-care deployment.	
ABN AMRO Bank / Capgemini — Software Engineer <i>Commercial banking platforms</i>	2021–2023 <i>India / Remote</i>
Designed Java services, REST APIs, tests, and Azure CI/CD workflows for digital onboarding and customer operations; supported changes associated with 60% higher conversion and 40% lower Customer Effort Score.	

Selected Technical Work

- Multimodal Alzheimer progression modeling:** fused T1 MRI, longitudinal cognitive scores, and APOE/CLU/BIN1 features using 3D CNN, GRU, MLP, and cross-modal attention encoders; achieved 0.918 ROC-AUC for stable versus progressive MCI.
- Promptable segmentation and sequence modeling:** repurposed a Stable Diffusion U-Net for deterministic one-step instance segmentation; built transformer models for anomaly prediction in HPC logs.

Research Output

11 peer-reviewed papers | first-author paper, AHFE 2026 | invited symposium, Gerontechnology 2026

- | | |
|---|--|
| <ul style="list-style-type: none">Evaluating Integrated Multimedia Communication Designed for Cognitive Accessibility <i>AHFE 2026. First author.</i>Streamlining Remote Care: Computer Access Solutions for Older Adults with Cognitive Impairment <i>AHFE 2026.</i>Embedding Self-Administered Dementia Monitoring into a Remote Caregiving Platform <i>AHFE 2026.</i>Bounded Delegation of a Large Language Model in a Conversational Agent for Dementia Support <i>IEEE AIXMHC 2026.</i> | <ul style="list-style-type: none">Only Speaking FACTS: Foundation Models for Activity-Centric Timeline Description <i>HumanSys 2026.</i>AlzGood: Validating Remote Care Systems in the Wild via an Interactive Deployment Framework <i>IEEE CHASE 2026.</i>AI-Care: A Conversational Agentic System for Task Coordination in Alzheimer’s Disease Care <i>IEEE CHASE 2026.</i>Promoting Autonomy in Older Adults with Cognitive Impairment <i>AHFE 2025.</i> |
|---|--|

Teaching, Recognition, and Skills

Teaching: UC Davis TA, computer science and technology ethics; 5/5 evaluation; Outstanding Graduate Teaching Assistant Award nominee.

Skills: Python, PyTorch, scikit-learn, FastAPI, Firebase, Java/Spring, TypeScript, React/Vue, SQL; multimodal ML, LLMs, computer vision, sensing, HCI evaluation.